

## DAVIS STATION, Antarctica

WMO: 895710

Lat: 68.574S

Lon: 77.967E

Elev: 23

StdP: 101.05

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | -31.7      | -29.5 | -36.8                         | 0.1 | -31.0 | -34.7 | 0.1 | -28.6 | 27.1                  | -7.0 | 24.6 | -7.1 | 2.9                   | 180  | 1.197 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |     |      |     |      | Evaporation WB/MCDB |      |     |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|-----|------|-----|------|---------------------|------|-----|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%  |      | 2%  |      | 0.4%                |      | 1%  |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB  | MCWB | DB  | MCWB | WB                  | MCDB | WB  | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e) | (f)  | (g) | (h)  | (i)                 | (j)  | (k) | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 3.2                    | 5.1             | 0.6  | 4.0 | 0.0  | 3.1 | -0.4 | 1.1                 | 4.1  | 0.5 | 3.2  | -0.1 | 2.5  | 7.9                  | 50   |

| Dehumidification DP/MCDB and HR |     |      |      |     |      |      |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|------|-----|------|------|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%   |     |      | 2%   |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP   | HR  | MCDB | DP   | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d)  | (e) | (f)  | (g)  | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| -1.0                            | 3.5 | 0.7  | -1.8 | 3.2 | 0.7  | -2.7 | 3.0 | 0.5  | 11.7          | 4.2  | 10.6 | 3.3  | 9.6  | 2.6  | 4.1               |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 22.3          | 19.1 | 15.7 | -32.9                      | 7.3 | 3.8                | 1.6 | -35.6                                              | 8.5 | -37.9      | 9.4 | -40.0      | 10.3 | -42.8      | 11.5 |    |    |
| (5)               |      |      | -33.0                      | 2.2 | 3.8                | 0.9 | -35.8                                              | 2.9 | -38.0      | 3.4 | -40.1      | 3.9  | -42.9      | 4.6  |    |    |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | -10.1 | 0.8   | -2.3  | -8.2  | -13.5 | -15.9 | -15.9 | -17.1 | -17.1 | -15.7 | -12.1 | -4.3  | 0.0  | (6)  |
| (7)  |                                                                                  | DBStd       | 8.20  | 1.72  | 2.36  | 4.08  | 5.43  | 6.09  | 5.85  | 6.81  | 6.02  | 5.50  | 4.46  | 3.31  | 1.94 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 7353  | 286   | 346   | 563   | 704   | 803   | 777   | 840   | 841   | 771   | 684   | 429   | 309  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 10395 | 544   | 579   | 822   | 954   | 1061  | 1027  | 1098  | 1099  | 1021  | 943   | 679   | 568  | (9)  |
| (10) |                                                                                  | CDD10.0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (10) |
| (11) |                                                                                  | CDD18.3     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 6.2   | 6.3   | 6.6   | 6.2   | 5.5   | 5.9   | 6.1   | 6.3   | 6.2   | 5.8   | 6.0   | 7.1   | 6.6  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 361   | 37    | 37    | 32    | 24    | 27    | 28    | 31    | 28    | 23    | 25    | 36    | 32   | (15) |
| (16) |                                                                                  | PrecMax     | 473   | 95    | 69    | 69    | 56    | 58    | 64    | 79    | 89    | 59    | 53    | 81    | 75   | (16) |
| (17) |                                                                                  | PrecMin     | 196   | 14    | 21    | 11    | 7     | 10    | 6     | 3     | 3     | 2     | 8     | 10    | 8    | (17) |
| (18) |                                                                                  | PrecStd     | 57    | 17    | 13    | 15    | 12    | 12    | 12    | 16    | 17    | 12    | 10    | 18    | 14   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 7.4   | 4.9   | 0.9   | -1.0  | -1.0  | -2.4  | -2.1  | -2.6  | -3.4  | -1.2  | 3.5   | 6.7  | (19) |
| (20) |                                                                                  |             | MCWB  | 1.8   | 0.6   | -2.0  | -3.3  | -3.6  | -4.3  | -4.2  | -4.2  | -4.9  | -4.2  | -0.6  | 1.3  | (20) |
| (21) |                                                                                  | 2%          | DB    | 5.4   | 2.8   | -0.8  | -2.8  | -3.8  | -4.1  | -4.1  | -5.1  | -4.7  | -2.9  | 2.1   | 5.0  | (21) |
| (22) |                                                                                  |             | MCWB  | 0.8   | -0.7  | -3.4  | -4.7  | -5.5  | -5.8  | -5.5  | -6.6  | -6.2  | -5.2  | -1.4  | 0.4  | (22) |
| (23) |                                                                                  | 5%          | DB    | 4.3   | 1.8   | -1.9  | -4.3  | -5.3  | -5.8  | -5.9  | -6.8  | -6.1  | -4.2  | 1.0   | 3.8  | (23) |
| (24) |                                                                                  |             | MCWB  | 0.2   | -1.4  | -4.2  | -5.9  | -6.8  | -7.3  | -7.2  | -8.1  | -7.7  | -6.2  | -2.0  | -0.1 | (24) |
| (25) |                                                                                  | 10%         | DB    | 3.4   | 0.9   | -2.8  | -6.0  | -7.1  | -7.3  | -7.4  | -8.4  | -7.6  | -5.5  | 0.0   | 2.8  | (25) |
| (26) |                                                                                  |             | MCWB  | -0.2  | -1.9  | -5.0  | -7.4  | -8.4  | -8.6  | -8.6  | -9.6  | -9.0  | -7.3  | -2.4  | -0.6 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 2.4   | 0.9   | -1.5  | -3.1  | -3.4  | -4.0  | -3.7  | -4.1  | -4.5  | -3.8  | 0.1   | 1.9  | (27) |
| (28) |                                                                                  |             | MDCB  | 6.5   | 3.7   | 0.5   | -1.5  | -1.6  | -2.7  | -2.6  | -2.8  | -3.7  | -2.0  | 1.7   | 5.4  | (28) |
| (29) |                                                                                  | 2%          | WB    | 1.3   | -0.1  | -3.0  | -4.5  | -5.3  | -5.7  | -5.4  | -6.5  | -6.2  | -5.0  | -0.8  | 1.0  | (29) |
| (30) |                                                                                  |             | MDCB  | 4.3   | 1.8   | -1.3  | -3.0  | -4.0  | -4.3  | -4.3  | -5.3  | -4.7  | -3.2  | 1.3   | 3.9  | (30) |
| (31) |                                                                                  | 5%          | WB    | 0.7   | -0.8  | -4.0  | -5.7  | -6.7  | -7.2  | -7.1  | -8.0  | -7.5  | -6.1  | -1.5  | 0.3  | (31) |
| (32) |                                                                                  |             | MDCB  | 3.5   | 1.2   | -2.2  | -4.3  | -5.4  | -5.9  | -6.0  | -6.8  | -6.1  | -4.3  | 0.6   | 3.1  | (32) |
| (33) |                                                                                  | 10%         | WB    | 0.2   | -1.5  | -4.9  | -7.4  | -8.4  | -8.5  | -8.6  | -9.5  | -9.0  | -7.3  | -2.3  | -0.2 | (33) |
| (34) |                                                                                  |             | MDCB  | 2.8   | 0.6   | -3.0  | -5.9  | -7.1  | -7.3  | -7.4  | -8.4  | -7.7  | -5.6  | -0.2  | 2.3  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 3.2   | 3.4   | 4.4   | 4.9   | 5.2   | 5.1   | 5.1   | 5.5   | 5.4   | 5.1   | 4.2   | 3.4  | (35) |
| (36) |                                                                                  |             | MCDBR | 4.3   | 3.9   | 3.9   | 4.4   | 5.3   | 5.2   | 4.1   | 5.8   | 5.0   | 5.1   | 4.6   | 4.3  | (36) |
| (37) |                                                                                  |             | MCWBR | 2.9   | 3.1   | 3.3   | 4.0   | 4.9   | 4.7   | 3.9   | 5.4   | 4.6   | 4.4   | 3.6   | 2.8  | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 3.8   | 3.4   | 3.9   | 4.5   | 5.3   | 5.3   | 3.8   | 5.8   | 5.2   | 5.1   | 4.3   | 3.8  | (38) |
| (39) |                                                                                  |             | MCWBR | 2.8   | 3.0   | 3.6   | 4.2   | 5.1   | 5.0   | 3.8   | 5.5   | 4.9   | 4.4   | 3.4   | 2.7  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.265 | 0.254 | 0.227 | 0.173 | 0.124 | N/A   | 0.127 | 0.168 | 0.209 | 0.223 | 0.249 | 0.266 | (40) |      |
| (41) |                                                                                  | taud        | 2.429 | 2.435 | 2.362 | 2.155 | 1.917 | N/A   | 1.894 | 2.108 | 2.263 | 2.282 | 2.327 | 2.374 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 990   | 942   | 861   | 665   | 158   | 0     | 117   | 652   | 866   | 970   | 1001  | 1003  | (42) |      |
| (43) |                                                                                  | Edh at Noon | 96    | 82    | 65    | 36    | 5     | 0     | 4     | 37    | 69    | 94    | 105   | 106   | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 7.06  | 4.45  | 2.30  | 0.75  | 0.07  | 0.00  | 0.01  | 0.40  | 1.83  | 4.28  | 6.81  | 8.02  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.42  | 0.35  | 0.15  | 0.04  | 0.01  | 0.00  | 0.00  | 0.03  | 0.12  | 0.18  | 0.35  | 0.44  | (45) |      |

## Historical Trends

|                             |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only           |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (2 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page